CYCLONE FREDDY FLOOD RISK MOZAMBIQUE (11 MARCH 2023)

Key Points

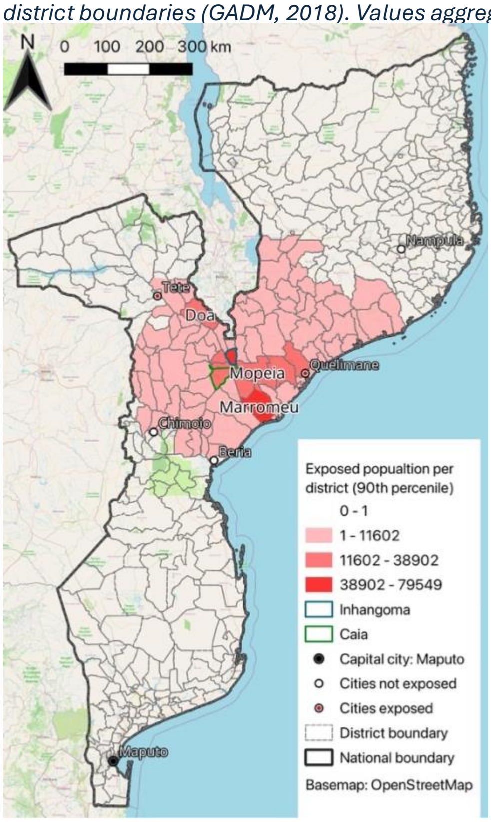
- Tropical Cyclone (TC) Freddy formed in the Indian Ocean on 4th February
- First landfall in Madagascar, followed by Mozambique
- Travelled back into Mozambique channel
- Expected to make second landfall on 11th March
- Flooding expected in 123 districts
- Marromeu District at highest risk
- Up to 478,482 people at risk of exposure (90th percentile)
- Road and rail infrastructure at risk

Flood Exposure by District

Table 1 *Five most exposed districts to flooding, outlining mean number of people exposed and 90th percentile.*

District	Population exposed (mean)	Population exposed (90th percentile)	District population
Marromeu	44,785	79,549	150,714
Inhangoma	26,186	63,616	103,460
Caia	14,698	38,901	85,303
Mopeia	26,428	36,299	148,266
Doa	12,662	26,012	255,680

Figure 1 *Number of people exposed to flooding, 90th percentile, per district atop of OpenStreetMap (2022) and district boundaries (GADM, 2018). Values aggregated with Natural Breaks (Jenks) (Copernicus, 2022; GloFAS,2022).*



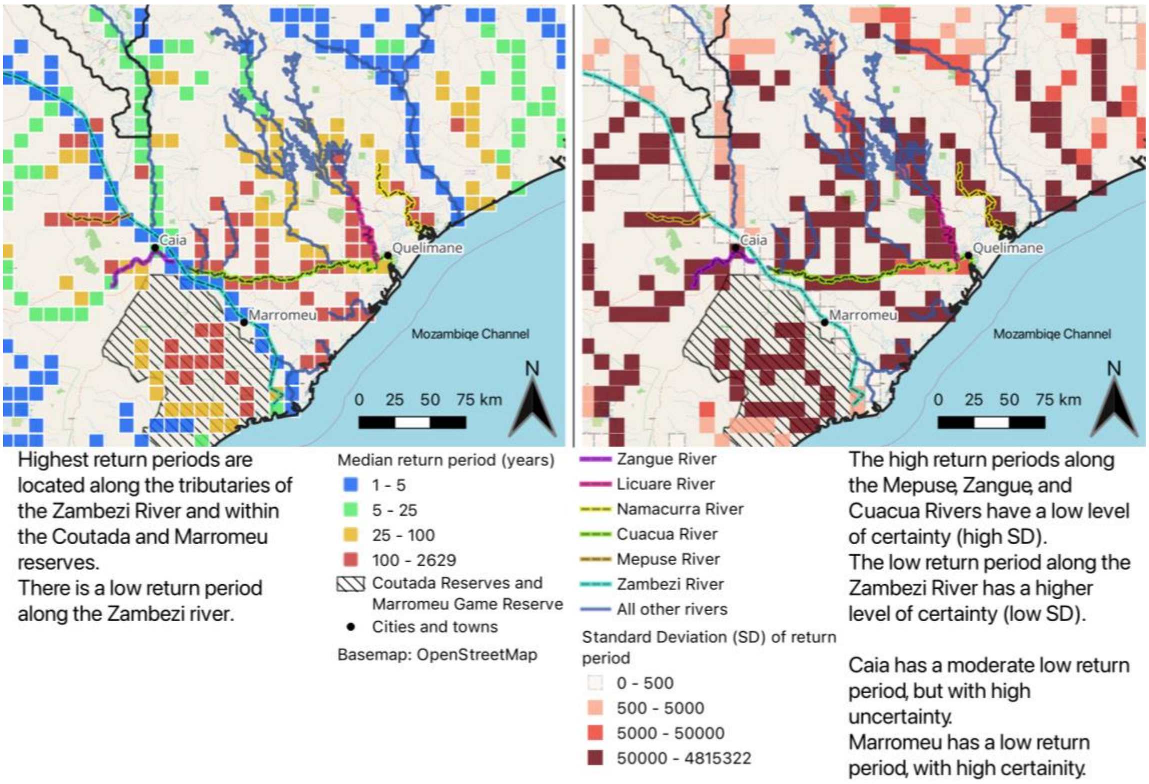
Up to 478,482 people are estimated to be at risk (90th percentile) nationally. Higher risk districts are in central Mozambique and along the Zambezi River basin and floodplains, creating underlying vulnerability. The five highest exposure districts, outlined in Table 1, are located in the basin. Marromeu is the most exposed district with an estimated worst case population exposure of 79,549, visualised in Figure 1. Figure 1 shows exposed population (90th percentile) per district. The five highest exposed districts from Table 1 are labelled in Figure 1.

Population Exposure

GloFAS forecasts, composed of 51 ensemble members, predict significant flooding across central

Mozambique. Forecast return periods range from 1 to 20,000 years across affected rivers. Given the extreme variation, median return periods are visualised in Figure 2. Figure 2 visualises return periods and their certainty in Mozambique's central coastal regions, including the highest risk regions of Marromeu, Inhangoma, Caia, and Mopeia labelled in Figure 1.

Figure 2 *Median and standard deviation of return periods, calculated from 51 ensemble members (Copernicus, 2022). Shown atop of OpenStreetMap (2022) and contextualised with rivers (Humanitarian Data Exchange 2020), locations of cities and towns, and reserves.*



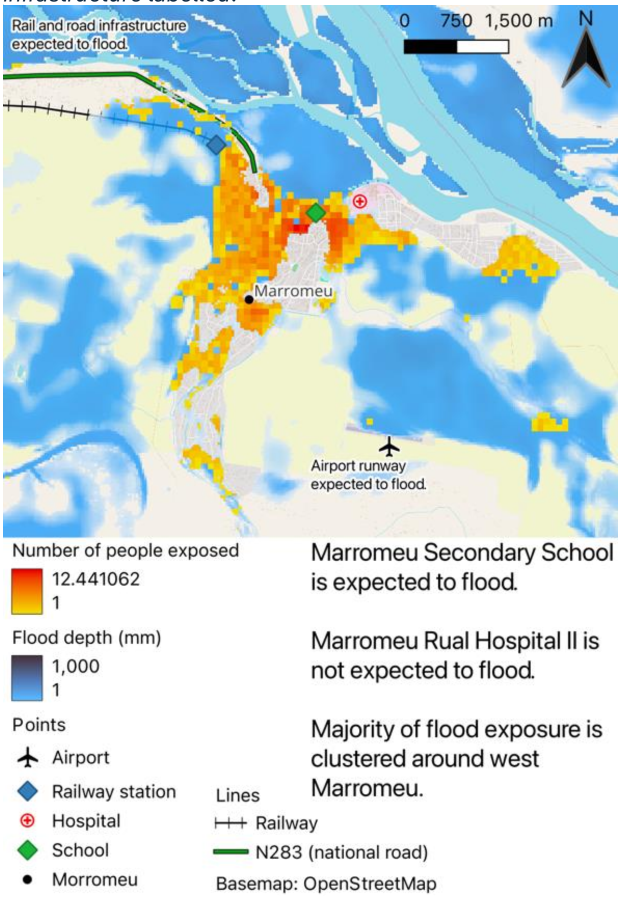
Highest return periods are located around the Zambezi River's tributaries, Namacurra River, and the floodplains and ephemeral streams of the Marromeu and Coutada reserves (Figure 2). The reserves have a low population and low exposure. Marromeu has the highest exposure.

Marromeu

- Predicted flooding is a 1 in 5 year event, with high certainty
- In this scenario an estimated 31,135 people are exposed to flooding in Marromeu

Figure 3 visualises exposed population in a 1 in 5 year event. Egress routes via road and rail are expected to experience flooding (Figure 3). The N283, a dirt road and the main exit route out of Marromeu, may deteriorate in condition from flood water. Similarly, the runway at Marromeu Airport is expected to flood with the potential to disrupt inbound and outbound flights. Marromeu Secondary School is expected to experience flooding. The Marromeu Rual Hospital II, 600m east of the secondary school, is not expected to flood. Unpopulated higher ground 1.5km west of the railway station and 500m west of the airport may provide refuge if transport links are rendered inviable (Figure 3).

__Figure 3__ Population exposure in Marromeu (WorldPop, 2020; GloFAS, 2022), atop of OpenStreetMap (2022) with key infrastructure labelled.



Vulnerability

Mozambique experiences widespread food insecurity, including higher flood risk areas which have an Integrated Food Security Phase (IPC) Classification of 2 (Stressed) (IPC, 2022). This vulnerability reduces adaptive capacity and ability to respond to flood events. This is reflected in Mozambique's low Human Development Index value of 0.490 (UNDP, 2022) which accounts for standard of living, life expectancy and education. The highest risk districts are rural and isolated, reducing access to emergency services and timely evacuation. There is a cholera outbreak in 29 districts, including Marromeu (WHO, 2023). The rapid growth in suspected cases, 5237 as of 19th February 2023 (WHO, 2023), presents a further risk as transmission may increase through contaminated water. These conditions have the ability to exacerbate potential impacts of flood events.

Limitations and Uncertainty

The population data, unlike the population itself, is static. Therefore, the population data cannot represent actual movement or growth in population. The population data provides estimates of size and distribution; it is limited temporally to its 2020 estimate. Exposed population numbers should not be treated as exact, instead they can indicate the general scope of exposure.

At approximately 10km, the resolution of GloFAS grid is course. Streams, flood pathways, and topographic features at small scales may not be represented at this resolution. Figures should be interpreted as an estimate, not exact, and subject to variation.

The return periods ranged from 1 to 20,000 years, suggest uncertainty within the model. Due to the large variation, median return periods (alongside SD) were visualised in Figure 2. Providing the range of return periods and distribution would be insightful, only median and SD were chosen as a balance between information depth and space constraints. Similarly to limitations of WorldPop and GloFAS outputs, the return periods are also estimates that have uncertainty.

Follow up question: Outline the methods used to map flood hazard in your bulletins and discuss the key limitations of the data.

The European Commission's Global Flood Awareness System (GloFAS) was used for predictions of river discharge for 11th March 2023. The GloFAS projection is comprised of 51 ensemble members, with each ensemble member providing different predictions of river discharge. Variation between ensemble members indicates uncertainty within the model, whereas ensemble members being broadly in consensus indicates a higher level of certainty. The 51 ensemble members are the output of weather forecast inputs into the LISFLOOD model with different forcings. LISFLOOD is a hydrological model that simulates river discharge (Knijff et al., 2010). At its core the bulletin relies on the probability of a flood occurring, it contains inherent uncertainty. Whilst this uncertainty can be communicated, the result is that the extent of inundation is an estimate and actual flooding may be more extensive than expected.

The bulletin aims to communicate risk, where risk is equal to hazard multiplied by exposure. To determine the extent of the hazard, Fathom Global Flood Model was used (Sampson et el., 2015). This flood model simulates flood inundation between 56°S and 60°N, including Mozambique, at a 30m resolution using the LISFLOOD-FP hydrological model. Using benchmark data from the UK and Canada, this model identifies between two thirds and three quarters of area at risk (Sampson et al., 2015). Whilst this model has a high predictive power for a global model applied to data scarce regions, underestimation of flood inundation is possible in Mozambique. Fathom produces flood hazard maps at different return periods; the appropriate layer was used matching the GloFAS estimated return period. The flood hazard layer was converted to make a binary raster of either 1 (flooded) or 0 (not flooded) using the reclassify by table tool in QGIS. Through clipping out of range values, a flood extent layer was made.

To determine exposure, the binary layer was multiplied by the values in the 2020 WorldPop gridded population data using the raster calculator tool in QGIS. As the flood layer was binary, following multiplication only the exposed population remained. The data was presented as zonal statistics (per district) and on a local scale (in Marommeu). Zonal statistics were calculated for all 51 ensemble members. After the attribute table of the zonal statistics was exported into R to calculate median and standard deviation, using the quantile() and sd() functions in base R, the updated attribute table was reimported into QGIS. The 90th percentile of people exposed per district (Figure 1) was composed with GADM3 boundaries and values in a graduated colour gradient. Zonal statistics can present general patterns on a national scale. A district with significant flood exposure within only a single city or the equivalent exposure spread across many cities would require different responses, this intra-district variation is unable to be captured with zonal statistics. To partially address this, the population exposure on a local scale in the most affected area (Marromeu in Figure 3) was made atop of an OpenStreetMap layer (a standard accessible basemap) with flood inundation and population exposure layers. Key infrastructure was labelled to assist emergency response and allocation of resources. Whilst providing actionable information, providing local scale flood forecasts for all areas of Mozambique is not feasible. This approach is consistent with established bulletin formats that include a national overview and local scale mapping of areas with the highest impact (Budimir et al., 2020).

The datasets used for this bulletin covered different years and different resolutions; a population dataset dated 2020 and with a resolution of 90m was used with a 2023 prediction 30m resolution flood model. The exposed population estimate is limited to the coarsest resolution, uniform distribution is assumed within the 90m pixel. If the boundary between flooded and not flooded 30m pixel is within a 90m population cell the entire 90m cell would be counted as exposed, this can be considered as overestimating population exposure. Populations are not fixed neither in space nor size thus a single static figure is unable to accurately represent a population. Given the longstanding food insecurity and presence of cholera (a highly communicable and fatal disease) it is plausible that changes to the population size has occurred within the 3 years since the 2020 WorldPop dataset.

Other limitations of the methods include the modelling of fluvial flooding only with the Fathom model, which does not include coastal or pluvial flooding. This is a significant limitation within the methods to not include forecasts of other flooding, given significant rainfall being associated with tropical cyclones. Research in urban situations has identified pluvial flooding can present equal economic risk as fluvial flooding, attributable to its wider inundation (Tanaka et al., 2020). Coastal regions of Mozambique with acute flood exposure are mostly low-lying and rural, not urban. The limited drainage infrastructure and low relief topography of the Zambezi River's floodplain may produce surface water storage to a similar extent.

Taken together, the flood hazard map provides key information with uncertainty acknowledged. The datasets used differ in both date and resolution, potentially reducing the accuracy of population distribution and quantification of exposure. Omission of other impacts, including other types of flooding, is a key limitation. Representation of the hazard data was subject to balance between national and local scale visualisations within constraints of bulletin space.

References

Budimir, M., Sneddon, A., Nelder, I., Brown, S., Donovan, A. and Speight, L., 2022. Development of forecast information for institutional decision-makers: landslides in India and cyclones in Mozambique. *Geoscience Communication*, *5*(2), pp.151-175.

European Union's Copernicus Global Flood Awareness System (GloFAS), 2022.

GADM, 2018. World / Mozambique / sub-divisions. Available at: https://gadm.org/maps/MOZ.html

Humanitarian Data Exchange, 2020. Available at: https://data.humdata.org/dataset/hotosm_moz_waterways

IPC (2022). Mozambique: Acute Food Insecurity Situation November 2021 - March 2022 and Projection for April - September 2022. Available at: https://www.ipcinfo.org/ipc-country-analysis/details-map/en/c/1155342/?iso3=MOZ

OpenStreetMap, 2022. www.openstreetmap.org/copyright

Sampson, C. C., A. M.Smith, P. D.Bates, J. C.Neal, L.Alfieri, and J. E.Freer (2015), A high-resolution global flood hazard model. Water Resour. Res., 51, 7358–7381, p.7358, doi:10.1002/2015WR016954

Tanaka, T., Kiyohara, K. and Tachikawa, Y., 2020. Comparison of fluvial and pluvial flood risk curves in urban cities derived from a large ensemble climate simulation dataset: A case study in Nagoya, Japan. *Journal of Hydrology*, *584*, p.124706.

UNDP (2022). Mozambique human development summary. Available at: https://hdr.undp.org/data-center/specific-country-data#/countries/MOZ

Van Der Knijff, J. M., Younis, J. and De Roo, A. P. J. (2010) 'LISFLOOD: a GIS-based distributed model for river basin scale water balance and flood simulation', *International Journal of Geographical Information Science*, 24(2), pp. 189–212. doi: 10.1080/13658810802549154.

WHO (2023). Cholera - Mozambique. Available at: https://www.who.int/emergencies/disease-outbreak-news/item/2023-DON443

WorldPop (www.worldpop.org - School of Geography and Environmental Science, University of Southampton; Department of Geography and Geosciences, University of Louisville; Departement de Geographie, Universite de Namur) and Center for International Earth Science Information Network (CIESIN), Columbia University (2018). Global High Resolution Population Denominators Project - Funded by The Bill and Melinda Gates Foundation (OPP1134076). https://dx.doi.org/10.5258/SOTON/WP00675